

Jeniya Sultana

Ph.D. Researcher, Wearable Sensing and Machine Learning for Construction Safety

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Summary

Ph.D. researcher combining a computer science foundation with civil and construction engineering research. I use wearable biosignals (EEG, EMG, PPG) and machine learning, including GAN-based data augmentation, to assess fatigue, cognitive load and safety in construction workers. Experienced in human-subjects research, NSF I-Corps customer discovery and teaching.

Education

Ph.D., Civil, Environmental and Construction Engineering <i>The University of Texas at El Paso</i>	Jan 2025 – Present El Paso, TX
M.S., Computer Science <i>Missouri State University</i>	Jul 2024 Springfield, MO
Graduate Certificate, Software Engineering <i>Missouri State University</i>	May 2024 Springfield, MO
B.Sc., Computer Science and Engineering <i>Chittagong University of Engineering and Technology (CUET)</i>	Dec 2019 Chittagong, Bangladesh

Research Experience

Graduate Research Associate <i>The University of Texas at El Paso</i>	Jan 2025 – Present El Paso, TX
- Design human-subjects experiments collecting EEG, EMG and PPG signals, and build ML pipelines to detect mental fatigue, cognitive load and physical strain in construction tasks. - Apply generative adversarial networks (GANs) to augment small, noisy biosignal datasets, improving model generalization. - Evaluate the physical and cognitive effects of passive exoskeletons on workers; results published at Construction Research Congress 2026.	
Entrepreneurial Lead, NSF I-Corps <i>National Science Foundation Innovation Corps</i>	Aug 2025 – Dec 2025 El Paso, TX
- Selected as graduate Entrepreneurial Lead to translate lab research toward real-world adoption. - Led customer discovery, conducting 50+ stakeholder interviews across industry to validate problem–solution fit. - Worked with faculty and industry mentors to assess feasibility, market need and adoption pathways; team secured competitive NSF I-Corps funding.	
Graduate Research Assistant (M.S. Thesis) <i>Missouri State University</i>	2022 – 2024 Springfield, MO
- Built context recognition and dynamic context detection frameworks for smart home voice commands using SBERT, attention, clustering and zero-shot learning; published at IEEE CCNC 2024 and 2025. - Designed metamorphic relations to test neural network robustness (IEEE CIoT 2024).	

Publications

- [1] **J. Sultana**, A. A. Raheem, J. Jang, J. S. Lee, Y. Kim, J. Kim, J. Kim, “EEG-Based Mental Fatigue Assessment during Exoskeleton-Assisted Construction Postures: A GAN-Augmented Approach,” *Construction Research Congress 2026*, ASCE, pp. 349–358, 2026. doi:10.1061/9780784486962.034
- [2] **J. Sultana**, R. Iqbal, “DyCoDe: Dynamic Context Detection from Voice Commands and Conversations in Smart Homes,” *IEEE 22nd Consumer Communications & Networking Conference (CCNC)*, 2025.
- [3] F. H. Athina, **J. Sultana**, D. D. Spandan, R. Iqbal, “Metamorphic Testing for Investigation of Context Recognition from Smart Home Voice Commands,” *7th Conference on Cloud and Internet of Things (CIoT)*, 2024.
- [4] **J. Sultana**, R. Iqbal, “A Framework for Context Recognition from Voice Commands and Conversations with Smart Assistants,” *IEEE 21st Consumer Communications & Networking Conference (CCNC)*, pp. 218–221, 2024.

- [5] A. Mondol, **J. Sultana**, M. Ahmed, A. B. Rashid, “Misinformation Detection in Cyber Smart Cities,” in *Cybersecurity for Smart Cities: Practices and Challenges*, Springer, pp. 105–120, 2023.
- [6] **J. Sultana**, “Towards Dynamic Context Detection from Voice Commands and Conversations with Smart Assistants in Smart Homes,” M.S. Thesis, Missouri State University, 2024.

Honors and Awards

Best Poster Presentation Award , Human Factors and Ergonomics Society (HFES) Regional Conference, El Paso, TX	Apr 2026
NSF I-Corps Team Funding , competitive award for research translation and customer discovery	2025
Bears Spotlight , featured by the Missouri State Computer Science department	2024
Graduate College Thesis Fund , Missouri State University, for proposal clarity and justification	Fall 2023
Best Undergraduate Thesis , recognized among the best theses in CSE, CUET	2019
Second Runner-Up (female category), National Collegiate Programming Contest	

Presentations

Poster: “Neuroergonomic Evaluation of Exoskeleton-Assisted Construction Postures Using Multimodal Physiological Monitoring”, HFES Regional Conference, El Paso, TX (<i>Best Poster Award</i>)	Apr 2026
Paper, Construction Research Congress (ASCE)	2026
Papers, IEEE CCNC (2024, 2025) and Conference on Cloud and Internet of Things (2024)	2024, 2025
Research presentations, 30th and 31st Frank Einhellig Graduate Interdisciplinary Forum, Missouri State University	2023, 2024

Professional Service and Affiliations

Session Chair , HFES Regional Conference, El Paso, TX	Apr 2026
Officer , Institute of Transportation Engineers (ITE) Student Chapter, UTEP	2025 – 2026

Teaching and Outreach

STEM Instructor <i>Discovery Center of Springfield</i>	Oct 2024 – Dec 2024 Springfield, MO
Designed and taught hands-on K–12 STEM programs in-house, at schools and through nationwide distance learning.	
Graduate Assistant, CODERS Project <i>Computer Science Opportunities, Development, and Education in Rural Schools, Missouri State University</i>	Aug 2022 – Jul 2024 Springfield, MO
Developed computer science curriculum and led workshops for rural K–12 teachers using Micro:bit, Scratch, robotics, smart home kits and drones; organized CS Olympiads.	
Graduate Assistant, ROSE Project <i>Research Opportunity for Smart Environments, Missouri State University</i>	Summer 2023 Springfield, MO
Taught STEM teachers Python and introductory machine learning; mentored teacher research projects.	
Lecturer <i>Bangladesh University of Textiles</i>	Nov 2021 – Aug 2022 Dhaka, Bangladesh
Taught computer programming to undergraduates across textile engineering departments.	

Certifications and Training

- Lean Six Sigma**: White Belt, Yellow Belt, Green Belt
- NSF I-Corps**: customer discovery and research translation (2025)
- Additional**: CITI Human Subjects Research; CCNA (IICT, CUET); Web and App Development; Ethical Hacking

Technical Skills

- Machine learning**: deep learning, GANs, transformers/SBERT, attention models, LSTM, transfer and zero-shot learning
- Sensing and signals**: EEG, EMG and PPG acquisition and processing; experimental protocol design
- Programming and tools**: Python, MATLAB, C/C++, Java, SQL, JavaScript/Node.js, PHP; OpenCV, MediaPipe, MySQL